

Programme : Diploma in Cloud Computing and Big Data	
Course Code : 5271	Course Title: Advanced Cloud Computing
Semester : 5	Credits: 4
Course Category: Programme Ccore Course	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To introduce advanced topics in cloud computing.
- To introduce Cloud Simulators
- To familiarize with Fundamentals of Cloud Security

Course Prerequisites:

Topic	Course Code	Course name	Semester
Basics of Cloud Computing	3273	Introduction to Cloud Computing	3
Cloud Computing Practical	3276	Cloud Computing Lab	3
Web Development Practices	4277	Web Application Development Lab	4

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Describe Different Cloud Enabling Technologies	14	Understanding
CO2	Explain Fundamental Cloud Architectures and Cloud Simulators	13	Understanding
CO3	Discuss DevOps, Docker Containers, and Infrastructure Automation on Cloud	15	Understanding

CO4	Summarize Fundamental Cloud Security and it's Mechanisms	16	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Describe Different Cloud Enabling Technologies		
M1.01	Summarize Broadband Networks and Internet Architecture	3	Understanding
M1.02	Explain Data Center Technology	3	Understanding
M1.03	Describe Virtualization Technology	4	Understanding
M1.04	Describe Multitenant Technology	2	Understanding
M1.05	Extend Service Technology	2	Understanding

Contents:

Broadband Networks and Internet Architecture- Internet Service Providers (ISPs), Connectionless Packet Switching (Datagram Networks), Router-Based Interconnectivity, Technical and Business Considerations.

Data Center Technology - Virtualization, Standardization and Modularity, Automation , Remote Operation and Management, High Availability, Security-Aware Design, Operation, and Management, Facilities, Computing Hardware, Storage Hardware, Network Hardware

Virtualization Technology- Hardware Independence, Server Consolidation, Resource Replication, Operating System-Based Virtualization, Hardware-Based Virtualization, Virtualization Management, Other Considerations

Multitenant Technology - Features, Common Characteristics

Service Technology - Web Services, REST Services, Service Agents, Service Middleware

CO2	Explain Fundamental Cloud Architectures and Cloud Simulators		
M2.01	Explain Fundamental Cloud Architectures	9	Understanding
M2.02	Discuss Cloud Simulators	4	Understanding
	Series Test – I	1	

Contents:

Fundamental Cloud Architectures - Workload Distribution Architecture, Resource Pooling Architecture, Dynamic Scalability Architecture, Elastic Resource Capacity Architecture, Service Load Balancing Architecture, Cloud Bursting Architecture, Elastic Disk Provisioning Architecture, Redundant Storage Architecture

Cloud Simulators - Introduction to CloudSim, Introduction to GreenCloud

CO3	Discuss DevOps, Docker Containers, and Infrastructure Automation on Cloud		
M3.01	Describe DevOps	5	Understanding
M3.02	Explain Containerization using Docker	5	Understanding
M3.03	Summarize Virtualisation with Docker Containers	2	Understanding
M3.04	Outline Infrastructure Automation on Cloud	3	Understanding

Contents :

DevOps - Introduction to DevOps, Benefits, DevOps Practices, Continuous Deployment, Continuous Delivery, Continuous Integration

Containerization using Docker - Docker, Containerization, Docker Architecture, Components of Docker, Docker Networks, Advantages of Docker

Virtualisation with Docker Containers

Infrastructure Automation on Cloud - Importance, Working, Benefits

CO4	Summarize Fundamental Cloud Security and it's Mechanisms		
M4.01	Summarize Basic Terms and Concepts related to Cloud Security	3	Understanding
M4.02	Outline Threat Agents	2	Understanding
M4.03	Describe Cloud Security Threats	5	Understanding
M4.04	Explain Fundamental Cloud Security Mechanisms	6	Understanding
	Series Test – II	1	

Contents:

Basic Terms and Concepts - Confidentiality, Integrity, Authenticity, Availability, Threat, Vulnerability, Risk, Security Controls, Security Mechanisms, Security Policies .

Threat Agents - Anonymous Attacker, Malicious Service Agent, Trusted Attacker, Malicious Insider

Cloud Security Threats : Cloud Security Threats - Traffic Eavesdropping, Malicious Intermediary, Denial of Service, Insufficient Authorization, Virtualization Attack, Overlapping Trust Boundaries, Flawed Implementations, Security Policy Disparity, Contracts, Risk Management .

Cloud Security Mechanisms- Encryption, Hashing, Digital Signature, Public Key Infrastructure (PKI), Identity and Access Management (IAM), Single Sign-On(SSO), Cloud Based Security Groups, Hardened Virtual Server Images.

Text / Reference

T/R	Book Title/Author
T1	Cloud Computing: Concepts, Technology & Architecture- Erl, Thomas, Ricardo Puttini, and Zaigham Mahmood, Pearson Education, 2013.
R1	Cloud Computing a Practical Approach - Anthony T.Velte , Toby J. Velte Robert Elsenpeter, TATA McGraw- Hill , New Delhi – 2010
R2	Mastering in Cloud Computing- Rajkumar Buyya, Christian Vecchiola and Thamari Selvi S, McGraw Hill Education (India) Private Limited, 2013.
R3	Cloud Computing: Theory and Practice- Dan C. Marinescu Elsevier Science, 2013, 1st Edition, Print Book ISBN :9780124046276, eBook ISBN :9780124046412
R4	Cloud Computing Fundamentals, Industry Approaches and Trends - Rishabh Sharma,, Wiley-2015

Online Resources

Sl.No	Website Link
1	https://www.znetlive.com/blog/virtualization-in-cloud-computing/
2	https://www.edureka.co/blog/devops-tutorial
3	https://aws.amazon.com/devops/what-is-devops/
4	https://www.simplilearn.com/tutorials/devops-tutorial/continuous-delivery-and-continuous-deployment#what_is_continuous_integration
5	https://www.geeksforgeeks.org/containerization-using-docker/
6	https://www.geeksforgeeks.org/virtualisation-with-docker-containers/
7	https://www.vmware.com/topics/glossary/content/infrastructure-automation.html